

Assignment 1

CFM

Select one of the index of your choice from the following list

- I. **NSE Nifty 50** (^NSEI)
- II. **BSE Sensex** (^BSESN)
- III. **IBEX 35** (^IBEX)
- IV. **OMXS 30** (^OMX)
- V. **ATX** (^ATX)

1. Verify if they follow the Semi-Annual schedule (Rebalancing Frequency). Identify the dominant sector in the index selected.
2. Which of these three European indices is most likely to move in sync with the Nifty 50? Justify based on sector similarities.
3. Give some of your observation from the data. For example: Look at the **2022 Global Slowdown**. Did the export-heavy Swedish market (^OMX) suffer more than the domestic-focused Indian market?
4. Reconstruct the exact composition of your selected International Index over a period of 15 years as follows
 - Create one Master Excel file for your selected index.
 - The file must cover the last **15 Years** broken down **Biannually** (every 6 months).
 - The file must contain **30 Sheets**. Each sheet must be named by its specific period (e.g., Sep-2010, Mar-2011) and contain the list of constituents that were part of the index during that specific window.
5. Write a generic Python Notebook that:
 - takes the number of years as input from user and give details of all constituents that were part of the selected index throughout this period.
 - Total number of stocks, Each Stock Name and Sector It belongs
 - Create a Master Database (by downloading Excel file for each selected stock) of these stocks.
 - Analyse which sector contributed the maximum number of survivors in this country?
6. **Corporate Action Tracking (Splits & Bonuses)**
 - **Focus:** Only on the stocks in the "**15-Year Legends**" list.
 - **Data Extraction:** For each of these survivors, gather details of all **Stock Splits** and **Bonus Issues** declared in the last 15 years.
 - **Format:** Tabulate the details including Event Type, Ratio, and Ex-Date.